

NICHOLAS TRIGGER

Biomedical Engineer

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SUMMARY

Biomedical Engineering alumnus from Duke University specializing in medical device R&D and design controls, with hands-on experience designing, prototyping, verifying, and validating hardware. Comfortable across embedded systems and benchtop testing under ISO/FDA-conscious development, with clinical research and leadership experience that bridges engineering and real-world clinical workflows.

EDUCATION

Duke University, Durham, NC

AUG 2022 – MAY 2026

Bachelor's of Science in Biomedical Engineering

Relevant Coursework: Biomaterials, Biomechanics, Medical Device Design, Embedded Medical Devices, Imaging Systems

EXPERIENCE

Research and Development Engineer

DEC 2025 – MAY 2026

Resolute Medical (Durham, NC)

- Designed and executed validation tests to support product evaluation against design requirements.
- Conducted comprehensive literature reviews to inform and refine experimental designs.
- Documented and analyzed experimental results, facilitating informed design decisions and technical reporting.

Classroom Research and Development Engineer

APR 2025 – MAY 2026

Duke University Biomedical Engineering. (Durham, NC)

- Designed, fabricated, and assembled electronic laboratory hardware for deployment across BME teaching labs.
- Developed a stimulus-responsive EKG simulator generating physiologically accurate signals for lab use.
- Validated hardware performance under extended use conditions to ensure durability and safety.

Clinical Research Associate

JUN 2023 – AUG 2024

Christus Spohn Health System (Corpus Christi, TX)

- Supported emergency medicine clinical studies by ensuring compliant data collection and thorough EMR review.
- Collected and documented data in accordance with IRB and FDA regulatory requirements, supporting compliant records for clinical investigations
- Collaborated effectively with physicians, residents, and research staff to facilitate ongoing clinical investigations.

PROJECTS

Central Line Associated Bloodstream Infection Prevention Device

nicholastrigger.com/projects/clabsi

- Developed handheld UV-C disinfection device for central-line hubs as Duke BME senior capstone, targeting 40,000 annual CLABSIs and eliminating dependency on nurse compliance with Scrub-the-Hub protocol.
- Validated device performance through design verification testing, achieving a 4-log CFU reduction and 2.49 mW optical output targeting the 263-273 nm germicidal window.
- Engineered three custom PCBs, an SLA & FDM 3D printed enclosure, and designed firmware to operate the device safely.

Radial Arterial Line Placement Training Device

nicholastrigger.com/projects/arm/

- Co-developed a durable, anatomically accurate arterial line placement training device to address the shortage of realistic, reusable simulation tools for radial arterial line insertion training
- Iterated through multiple prototype revisions incorporating clinician/user feedback to improve pulse simulation fidelity, tissue realism, and durability under repeated needle insertions

CERTIFICATIONS AND SKILLS

Certifications: CSSC: Lean Six Sigma White Belt, Nordic Semi: nRF Connect SDK Fundamentals, Matter Fundamentals

Technical Skills: BioSignal Processing, X-ray, MRI, CT, PET, Medical Device Prototyping, Biomechanics, ISO Standards, FDA Regulations, EMG, Medical Image Processing, Medical Firmware, GD&T, PCB Layout, PCB Design, Python, C, C++, Embedded C, FreeRTOS, Zephyr, JavaScript, TypeScript, HTML/CSS, SQL, SPI, I2C, UART, CAN, BLE, USB, V&V, Bench Testing, System Validation, Bacterial Testing, Phantoms, Data Collection & Analysis, PCB Triage, Prototyping

Tools: SolidWorks, Fusion 360, AutoCAD, OnShape, SolidWorks, KiCAD, EAGLE CAD, Git, GitHub, GitLab, 3D Printing, Laser Cutting, CNC Milling, Manual Machining, Waterjet Cutting, PCB Assembly, PCB Fabrication, Microsoft Office, LaTeX